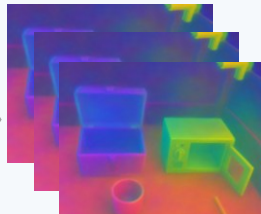
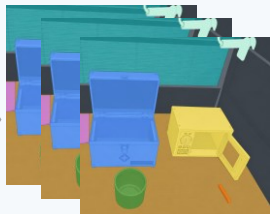
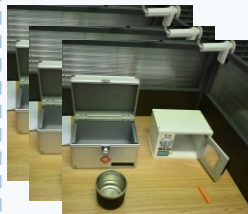


(a) Object-Background Decoupling

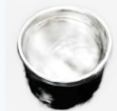
Multi-view RGB

Instance Masks

Object Ownership



Object 3DGS in Local Frames

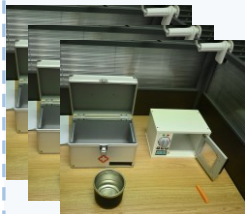


Background-aware 3DGS

Multi-view RGB

Background Completion

Background 3DGS

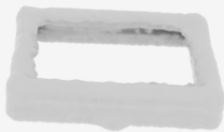
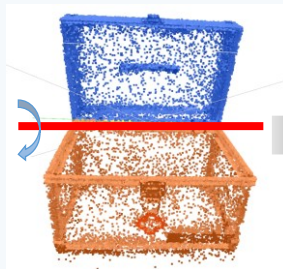


(b) Hinge-Parameterized Operable Asset

Axis, Center,
Range

Moving/Fixed Partition

Part-wise
Proxies



3DGS + Joint + Collision

Articulated Scene Asset

